Purchasing & Supplier Pricing Analysis

Tools: Power BI, DAX
Dataset: Simulated purchasing transactions
Reporting period: January–February 2026

1. Executive Summary

This project analyzes purchasing transactions to understand spending patterns, monitor purchase status, compare supplier prices across material categories, and estimate potential cost-saving opportunities.

An interactive Power BI dashboard was developed to present purchasing KPIs, material spending, supplier price comparisons, and estimated savings. The analysis recorded **₱150,910 in completed purchases**, **₱21,300 in pending purchases**, and an estimated **₱8,822.50 in potential savings** based on differences in supplier average prices.

The findings demonstrate how purchasing data can support supplier evaluation and help identify opportunities for further cost review. The dataset is simulated, and the savings figure is an analytical estimate rather than realized savings.

2. Business Objective

The objective was to explore how purchasing data can help answer the following business questions:

- How much has been spent on completed purchases, and how much remains pending?
- Which material categories account for the greatest purchasing expenditure?
- How do average unit prices differ among suppliers for the same material?
- What potential savings could be investigated by benchmarking supplier prices?

3. Dataset and Methodology

The dataset contains 15 simulated purchasing records covering January and February 2026. Each record includes a purchase date, supplier, material, quantity, unit price, and purchase status.

The data was analyzed using Power BI and DAX. Measures were developed to calculate completed purchasing spend, pending spend, completed purchase counts, average unit prices, and estimated potential savings.

Supplier prices were compared **within each material category** to avoid misleading comparisons between suppliers selling different materials. The savings estimate uses the lowest recorded average unit price for each material as a benchmark and applies the price difference to quantities purchased from higher-priced suppliers.

4. Key Findings

Purchasing expenditure and status

- **Completed purchasing spend:** ₱150,910

- **Pending purchasing value:** ₱21,300
- **Completed purchase records:** 13 out of 15
- **Estimated potential savings:** ₱8,822.50

Completed purchases represent the majority of the recorded purchasing value. Pending transactions are worth ₱21,300 and could be reviewed to confirm their current status and expected completion.

Spending by material

Completed purchasing expenditure was distributed as follows:

Material	Completed spend
LED Panel	₱94,910
Downlight	₱33,475
Driver	₱22,525
Total	**₱150,910**

LED Panels accounted for the largest share of completed purchasing expenditure, at approximately 62.9%. This makes LED Panel pricing a useful priority for further supplier negotiations and cost-control reviews.

Supplier price comparison

The lowest recorded average unit price differed by material:

Material	Lowest average price	Supplier
LED Panel	₱817.50	Metro Electrical
Downlight	₱295.00	Metro Electrical
Driver	₱170.00	Metro Electrical

Within this sample, Metro Electrical had the lowest average recorded price in all three material categories. However, price alone does not establish the best supplier; product specifications, quality, delivery performance, availability, and purchasing terms must also be considered.

Estimated savings opportunity

The analysis estimated a total potential savings opportunity of **₱8,822.50** by comparing each supplier's average price with the lowest average price recorded for the same material.

Supplier	Estimated opportunity
Bright Source	₱5,937.50
Alpha Supplies	₱2,885.00
Metro Electrical	₱0.00
Total	**₱8,822.50**

Bright Source accounted for the largest estimated opportunity, followed by Alpha Supplies. These figures indicate where a purchasing team could investigate pricing differences; they do not guarantee that equivalent products could be purchased at the benchmark prices for all quantities.

5. Recommendations

1. **Review LED Panel purchasing costs.** Because LED Panels account for the largest share of completed spend, validate supplier quotes and investigate opportunities to negotiate more competitive pricing.
2. **Review higher-priced supplier quotations.** Start with Bright Source and Alpha Supplies, comparing equivalent products and specifications before negotiating.
3. **Validate supplier comparability.** Before switching suppliers, assess product quality, technical compliance, delivery reliability, minimum order quantities, and payment terms.
4. **Monitor pending purchases.** Review the two pending transactions to confirm whether they require follow-up or status updates.
5. **Continue tracking purchasing KPIs.** Refresh the analysis with actual transaction data over a longer period to monitor price trends and evaluate whether negotiated savings are realized.

6. Limitations

This project uses a small, simulated dataset created for portfolio and learning purposes. It does not represent actual company purchasing activity or real supplier quotations.

The estimated savings calculation assumes that the lowest recorded average unit price for a material could serve as a benchmark for higher-priced purchases. It does not independently account for differences in product specifications, quality, supplier capacity, shipping costs, discounts, or contract terms. Therefore, the savings figure should be interpreted as an indicative opportunity for further investigation, not a confirmed financial result.

The limited number of transactions and short reporting period also prevent reliable conclusions about long-term pricing trends or supplier performance.

7. Conclusion

This project demonstrates how Power BI and DAX can be used to transform purchasing transactions into an interactive dashboard for expenditure monitoring, supplier price comparison, and opportunity identification.

The analysis highlights LED Panels as the largest spending category and identifies supplier price differences worth investigating. With actual purchasing records and additional supplier-performance data, the same approach could support more informed procurement decisions, cost-control initiatives, and ongoing performance monitoring.